Task 2.4 - DRAFT scientific evidence

Research question: how quickly can the buffer reveal the stored memory bit while limiting unwanted logical transitions, leakage and emission after the pulse? This report is regenerated from finished jobs. It is intentionally a draft until numerical and physical review are complete.

Execution snapshot: 104 completed, 2 timeout. Completed means the job returned saved outputs. It does not mean converged, optimized, or accepted.

Task 2.4(a): derive the circuit interaction and physical memory drive. Task 2.4(b): measure output information and stored-bit disturbance separately. Task 2.4(c): compare shaped controls with tuned constant drives under common constraints. Task 2.4(d): validate an explanation against cutoffs, solvers and circuit limitations.

Available evidence: 83 completed pulse jobs, 14 spectral jobs and 6 coherence-control jobs. Discovery uses reduced cutoffs; larger-cutoff and tighter-solver results are separate rows. Missing jobs remain pending or explicitly failed/timed out.

Headline status: no global optimum, actual single-shot fidelity, globally certified switching rate, or validated improvement is claimed by this automated document. A final result requires matched comparisons and independent scientific acceptance.

The CSV, Pareto JSON, job manifest, source hashes, complete status list and executable evidence notebook accompany this PDF. Figures show only saved outputs; empty figures mean evidence has not yet completed.

2.4(a) - circuit-to-readout mechanism

At the specified ATS bias, the first pump-order potential is +2 EJ epsilon_p cos(omega_p t) sin(Phi). Its cubic two-memory/one-buffer resonant term has a negative coefficient. A pi pump-phase change gives the repository's positive-g2 convention, with the buffer-drive phase chosen consistently.

$$g_2 = E_J \varepsilon_p \varphi_a^2 \varphi_b / 2, \quad \omega_p = 2\omega_a - \omega_b$$

The memory readout drive is a separately coupled microwave force, not an automatic extra term from the two-photon ATS pump. For the pulse module's phase convention the rotating-frame drive and initial displacement are:

$$H_Z = \varepsilon_Z(t)(e^{i\phi} a^\dagger + e^{-i\phi} a), \quad \dot{A}|_Z = -i\varepsilon_Z e^{i\phi}$$

The buffer pump and cubic exchange cancel on |+alpha,0> and |-alpha,0> in the lossless ideal model. Weak real readout drive moves the memory initially along its imaginary direction; the local two-photon response then produces opposite real buffer fields. Neglecting the small shared loss-induced background:

$$\langle b \rangle_{\pm} \simeq \mp \varepsilon_Z / (2\alpha g_2)$$

Higher phase orders modify exchange matrix elements. Recalibrating the buffer drive to cancel the mean force does NOT restore exact coherent-state darkness. Every corrected model therefore needs a matched no-readout evolution. The pulse jobs include such photon, logical-population and leakage baselines.

First-order RWA does not produce arbitrary static Kerr terms. The sine model retains selected-channel phase corrections; circuit_correction=True additionally includes the audited compensated dispersive diagonal through the chosen phase order. These remain effective Hamiltonians, excluding a full GHz circuit and bath-dressing simulation.

Primary sources: supplied challenge Task 2.4; Reglade et al., arXiv:2307.06617v2, Supplement S3-S4 and S8.5. The independent audit is included as physics_audit.md where available.

2.4(b) - information is not a bit flip

Useful measurement, logical dephasing and unwanted logical population transfer are different effects. The primary reference is the full two-mode Lindblad model, retaining both supplied losses. The fast-buffer approximation is not uniformly valid at g2/kappa_b about 0.293 and epsilon_d/kappa_b about 1.295.

$$dY_s = 2\sqrt{\eta\kappa_b}\,\mathrm{Re}(e^{-i\phi}B_s)dt + dW$$

$$\mathcal{I}_\phi = \eta\kappa_b\int[\mathrm{Re}(e^{-i\phi}\Delta B)]^2 dt, \quad \mathrm{SNR}^2 = 4\mathcal{I}_\phi$$

This matched-filter quantity assumes unit white measurement noise. Under coherent outputs and equal Gaussian covariance, I=1 would give 15.9% error; mean fields alone do not establish that distribution or a measured fidelity. The nominal real quadrature and optimized fixed quadrature are recorded separately, so phase recalibration is explicit. A time-varying local oscillator is not silently assumed.

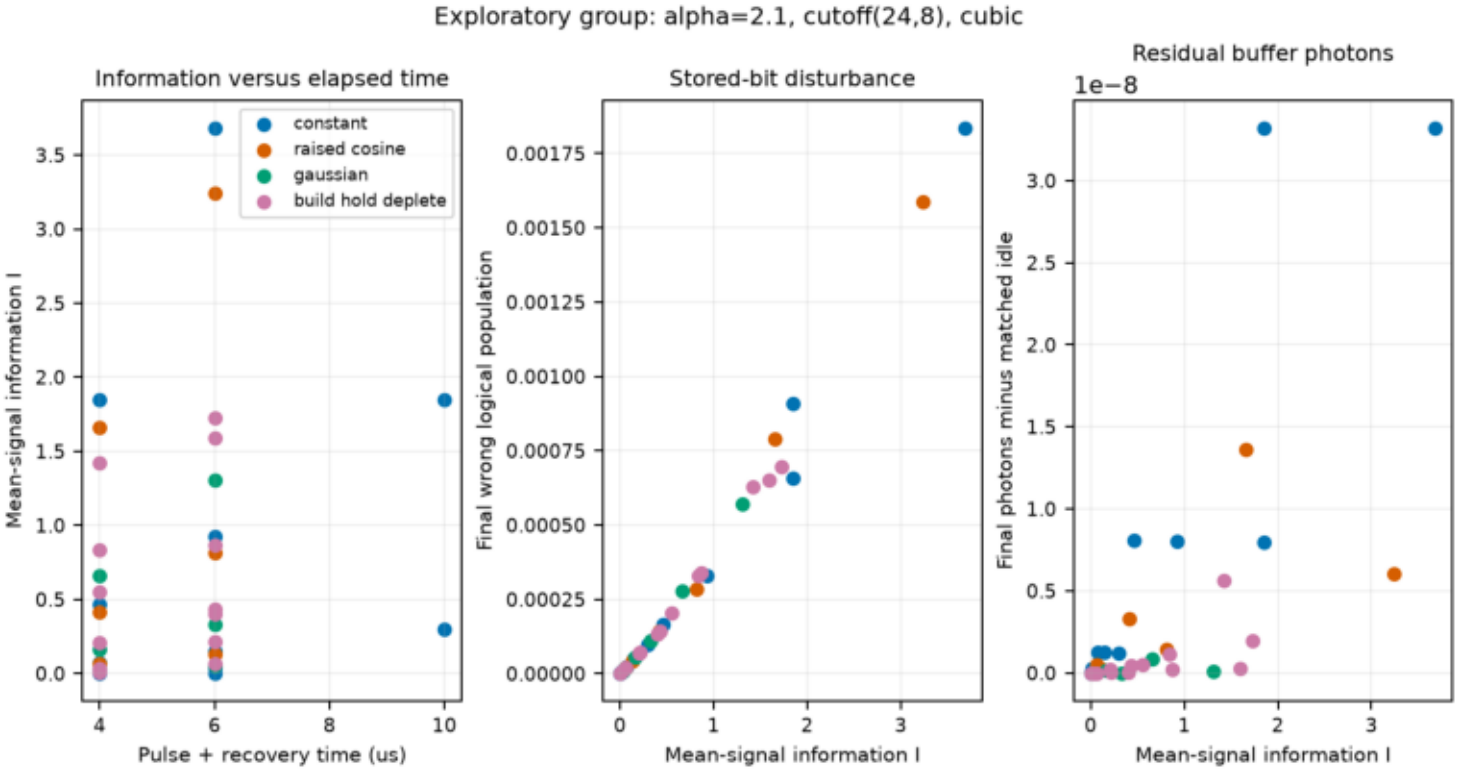
$$P_{\mathrm{code}} = |C_+\rangle\langle C_+| + |C_-\rangle\langle C_-|$$

Logical populations use the orthonormal sums/differences of normalized even and odd cats. The coherent inputs have a small finite-alpha wrong-logical offset, which is retained. Final wrong population, leakage and matched-idle-subtracted excesses are distinct; signed differences are not clipped or reinterpreted as probabilities.

Pulse-end information excludes later emission. Recovery photons and the information emitted after control termination are separately saved. A small mean buffer field does not certify a small total photon population: incoherent photons must also be counted.

A returned Liouvillian mode is not automatically the physical switching rate. Relevant overlaps, left/right residuals, conjugate pairs, time-domain behavior, cutoff convergence and the validity of a two-state model all require review.

2.4(c) - executed control trade-offs



Each point is a completed full two-mode pulse calculation. The largest completed group with identical physical settings, cutoffs and solver tolerances is shown; its amplitude and duration vary. All other groups remain in the CSV/JSON, with Pareto screening performed separately within each group. Compare equal-duration and peak-bound controls before judging shape effects. Apparent Pareto membership does not certify an advantage. Idle controls and signed background-subtracted residual photons are retained. The total clock includes the stated recovery interval.

2.4(c) - matched reference comparisons

Matched comparisons below use the same elapsed time, peak bound, physical settings and cutoffs. Ratios are observed numerical metrics, not accepted improvements. Missing matched controls are explicitly unproven.

raised_cosine, drive/g2=0.1, T=2 us, [24, 8]: I/I_constant=0.89658; delta wrong=-3.01e-06; delta leakage=-4.61e-10.

raised_cosine, drive/g2=0.25, T=2 us, [24, 8]: I/I_constant=0.89663; delta wrong=-2.1e-05; delta leakage=-2.94e-09.

raised_cosine, drive/g2=0.5, T=2 us, [24, 8]: I/I_constant=0.89685; delta wrong=-0.00012; delta leakage=-1.2e-08.

gaussian, drive/g2=0.1, T=2 us, [24, 8]: I/I_constant=0.35616; delta wrong=-1.57e-05; delta leakage=-8.45e-10.

gaussian, drive/g2=0.25, T=2 us, [24, 8]: I/I_constant=0.35628; delta wrong=-0.000109; delta leakage=-5.24e-09.

gaussian, drive/g2=0.5, T=2 us, [24, 8]: I/I_constant=0.35682; delta wrong=-0.000633; delta leakage=-2.14e-08.

50 further comparisons are retained in summary.json and the notebook.

A constant-drive amplitude sweep is the reference, not only the weak Task 2.3 drive. Gaussian and smooth-edge pulses pay for their turn-on/off intervals. Signed depletion is useful only if its reduced residual excitation compensates its information or time cost.

2.4(d) - validation and remaining limits

Successful execution is not convergence. The following comparisons record available differences; no automated physical acceptance is assigned. Separate Na/Nb increases and tighter time/solver resolution remain necessary for headline claims.

constant, drive/g2=0.5, T=4 us; [24, 8] versus [40, 12]: deltaI/I=1.21e-05; absolute delta wrong=8.61e-06; delta leakage=7.76e-09.

constant, drive/g2=0.5, T=4 us; [32, 12] versus [40, 12]: deltaI/I=7.89e-08; absolute delta wrong=3.6e-08; delta leakage=4e-11.

constant, drive/g2=0.5, T=4 us; [32, 14] versus [40, 12]: deltaI/I=3.45e-07; absolute delta wrong=2.37e-07; delta leakage=6.41e-11.

constant, drive/g2=0.5, T=4 us; [32, 12] versus [40, 12]: deltaI/I=8.16e-08; absolute delta wrong=3.6e-08; delta leakage=3.93e-11.

raised_cosine, drive/g2=0.5, T=4 us; [24, 8] versus [40, 12]: deltaI/I=1.02e-05; absolute delta wrong=7.45e-06; delta leakage=7.51e-09.

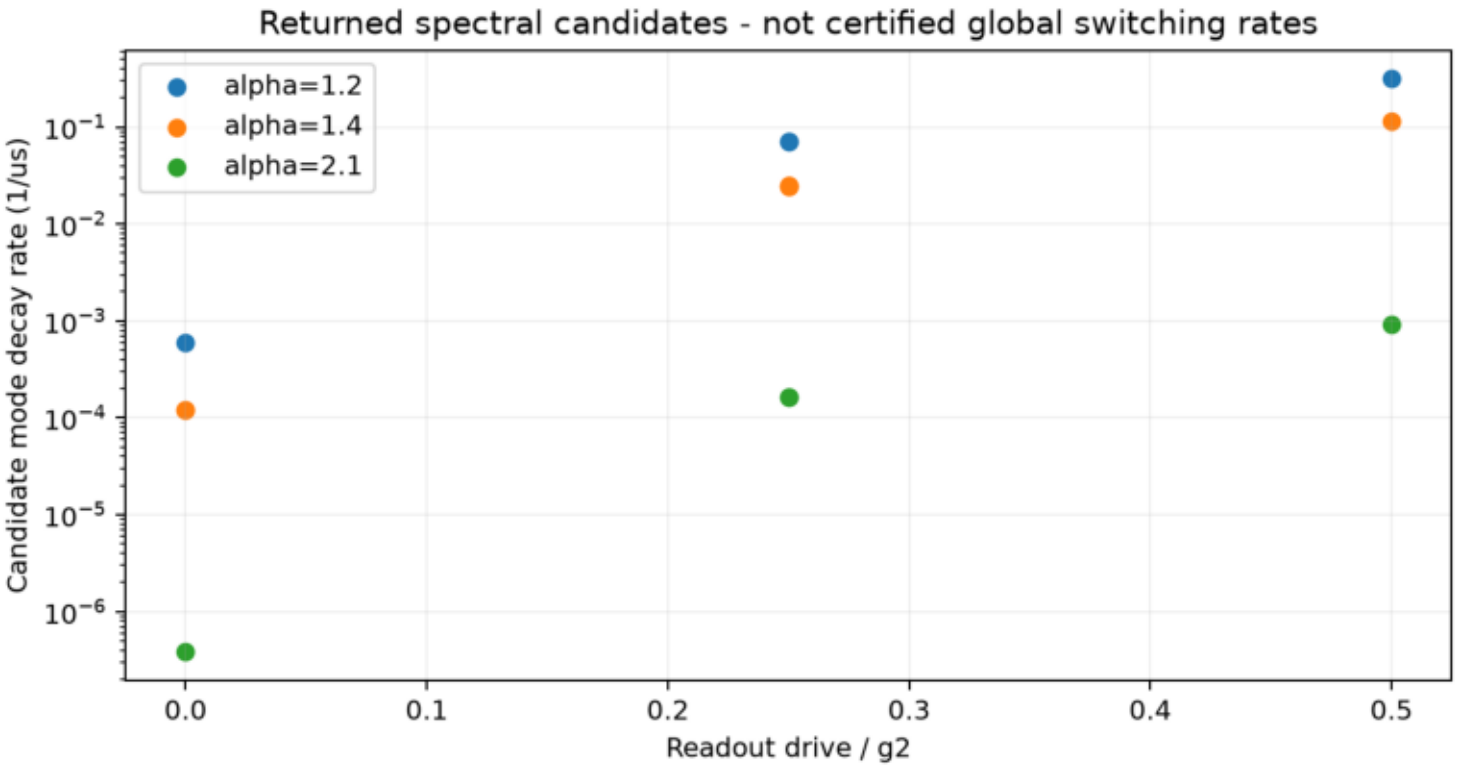
raised_cosine, drive/g2=0.5, T=4 us; [32, 12] versus [40, 12]: deltaI/I=6.63e-08; absolute delta wrong=3.09e-08; delta leakage=1.8e-11.

7 additional comparisons are retained in summary.json and the notebook.

Discovery at 24x8 is not final physical-target evidence. Boundary populations, trace and signed eigenvalue residuals are retained, but small boundary occupancy alone does not certify an exponentially small switching rate.

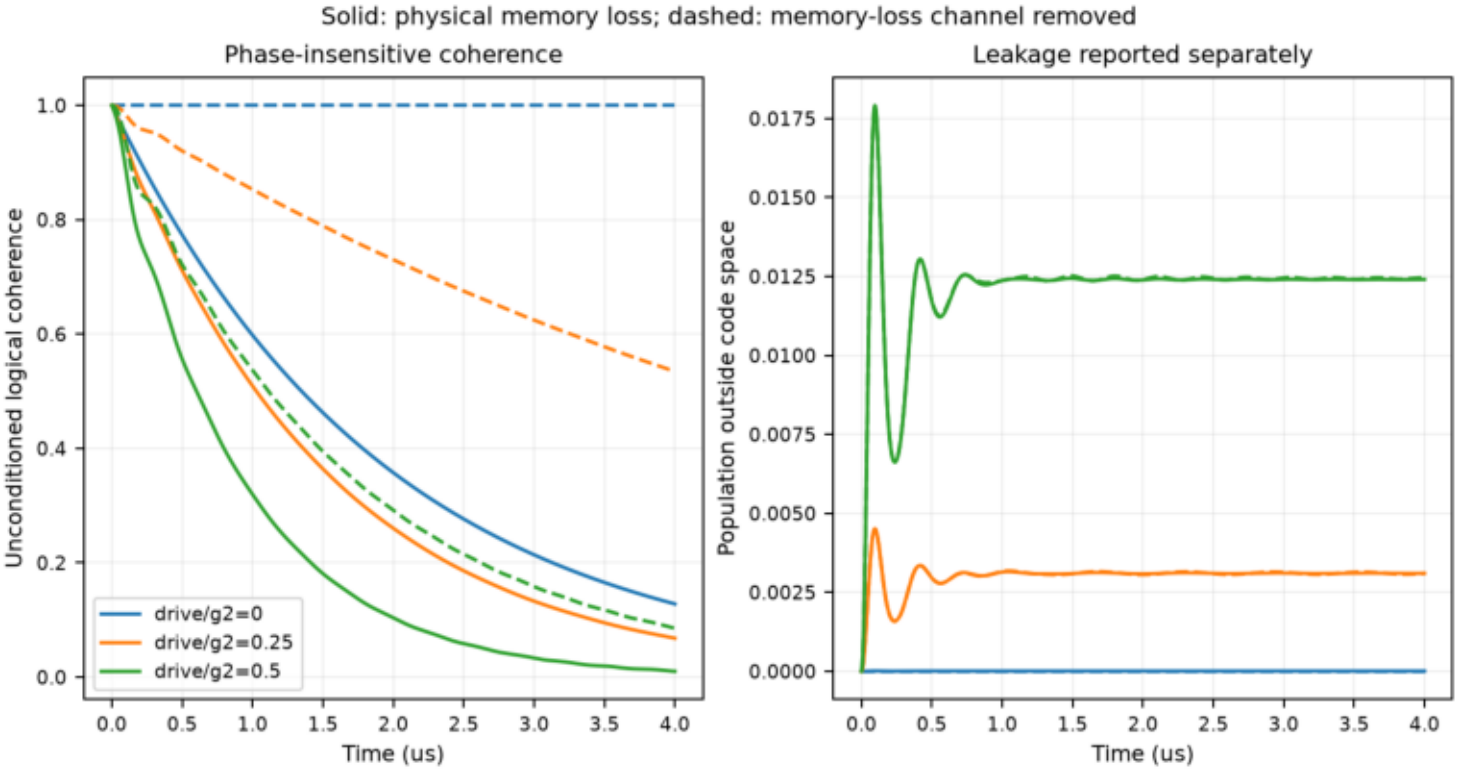
Robustness distinguishes fixed nominal microwave/LO calibration from recalibration. Phase/amplitude tests, no-memory-loss controls and sine-exchange cases are separate experiments. A Hamiltonian-only correction omits bath dressing, flux noise and fabrication uncertainty.

2.4(b,d) - candidate switching modes



Only numerically resolved returned candidates are plotted. All case statuses and unresolved values remain in summary.json. A near-shift eigensystem is not a completeness or global-dominance proof. Complex modes or several visible contributions can invalidate a single exponential two-state interpretation. Time-domain diagnostics, shift/cutoff changes and the one-way-versus-imbalance factor of two must be reviewed separately.

2.4(b,d) - dephasing and leakage controls



An initial even cat probes transverse logical coherence. The magnitude sqrt(X²+Y²) removes a common deterministic Bloch phase; it does not remove leakage, shown separately. Reduced-memory coherence can also decrease from reversible entanglement with an occupied buffer. It is therefore not a pure-dephasing-rate measurement or simply the overlap of emitted pointer fields. Code-conditioned coherence is a diagnostic, not a postselected fidelity. Memory-loss-off and zero-drive controls help separate physical contributions; logical switching remains an independent measurement. No fitted dephasing rate or single-shot fidelity is inferred.

Provenance, limitations and review

Every plotted value comes from a completed manifest job and its saved output. Source hashes and job IDs are retained in summary.json. Failed, timed-out and pending cases are not replaced with predictions or extrapolated measurements.

This automated report stays DRAFT until a scientific reviewer checks signs and 2pi factors, circuit resonance conditions, logical definitions, measurement-noise normalization, rate identification, truncation effects and fairness of each proposed improvement.

No actual single-shot fidelity, globally optimal pulse, protected long-time physical-target rate, exact full-circuit simulation or completed final submission is asserted. A negative optimization finding is acceptable if supported by matched numerical evidence.

AI assistance: Codex agents assisted with circuit and numerical audits, software, experiment design and report generation. Numerical findings must be supported by executed calculations. Independent validation and unresolved disagreements are retained in the accompanying records.

References: DuQuantum 2026 Alice & Bob challenge Task 2.4; Reglade et al., arxiv.org/abs/2307.06617; Dubovitskii, arxiv.org/abs/2407.17299. The latter's adiabatic expressions are qualitative checks unless validated against this full two-mode model.

Rebuild this draft from the campaign snapshot: python -m tasks.task_2_4_campaign_report --output <campaign-folder>. The executable notebook reads the report directory without rerunning simulations. CSV and JSON accompany all figure exports.